

MANIT SARKAR

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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at Thapar Institute of Engineering and Technology, with a **CGPA of 8.95/10** and a **merit scholarship**. Interests span Mechanical Engineering (CAD/Materials), Astrophysics (Gravitational/Extragalactic), and Machine Learning (Computer Vision/Neural Networks), with hands-on experience in ML-driven data pipelines and fatigue-based failure analysis. Proficient in Python ML frameworks (TensorFlow, PyTorch, Scikit-learn) and simulation tools (CREO, MATLAB). Actively researching multiwavelength imaging and satellite trajectory prediction.

EDUCATION

Thapar Institute of Engineering and Technology (TIET)

Patiala, India

B.E. Mechanical Engineering (7th Semester, Final Year) — CGPA: **8.95 / 10**

Expected: June 2027

- **Merit Scholarship** — **50% tuition fee waiver** for academic performance in 2nd year of B.Tech.

INTERNSHIP

Summer Internship

June – July 2026

Maruti Suzuki India Limited | Mentor: Gaurav Gupta, Manager (*certificate*)

- Interned with MSIL's Powertrain Testing & Validation team (Gurugram) and gained exposure to the working and testing of engine components and assemblies.
- Authored an independent study, "*Failure Mode Analysis of Engine Components*", applying FMEA and fatigue analysis (Arai's stress concentration factors, Goodman & Coffin-Manson criteria, S-N curves) to crankshaft/piston failures.

Experiential Learning (EL) Summer Internship

June – July 2025

Thapar Institute of Engineering and Technology | Mentor: *Dr. Mamta Gulati*

- Classified galaxy morphology (spiral vs. elliptical vs. irregular) using image-based and photometric (brightness, magnitude, star formation rate) inputs.
- Applied supervised ML models — decision trees, random forests, gradient boosting; **XGBoost achieved ~95% accuracy**.

RESEARCH & PROJECTS

Multiwavelength Galaxy Morphology Classification using AI/ML

Ongoing

Student Research | Thapar Institute of Engineering and Technology | Mentor: *Dr. Mamta Gulati*

- Extending EL internship work with a **CNN-based pipeline** to classify galaxies using multiwavelength (3+ bands) imaging data, moving beyond the prior numerical/tabular approach.

Flow-Induced Vibration Wind Energy Harvester

Ongoing

B.E. Capstone Project | Thapar Institute of Engineering and Technology | Mentor: *Dr. Ashish Purohit*

- Designing and analysing a harvester that converts **flow-induced vibrations into electrical energy**.

CERTIFICATIONS

Machine Learning Specialization (3 courses) (*certificates*)

June 2024

DeepLearning.AI & Stanford University | Andrew Ng | Coursera

- Supervised & Unsupervised Learning, Recommender Systems, Reinforcement Learning, Advanced Learning Algorithms

TensorFlow Developer Certificate Courses (4 courses)

Jul – Aug 2024

DeepLearning.AI | Laurence Moroney / Andrew Ng | Coursera

- Intro to TensorFlow for AI/ML/DL (Jul 2024) | CNNs in TensorFlow (Jul 2024) | NLP in TensorFlow (Aug 2024)

Deep Learning Specialization (1 course)

Sep 2024

DeepLearning.AI & Stanford University | Andrew Ng | Coursera

- Neural Networks & Deep Learning

TECHNICAL SKILLS

Programming	Python (primary), C / C++
ML / AI	Supervised & Unsupervised Learning, Computer Vision, Neural Networks, NLP, Image Segmentation
Libraries	TensorFlow, PyTorch, Scikit-learn, NumPy, Matplotlib, OpenCV, MediaPipe, YOLO, Astropy, AstroQuery, SpiceyPy
CAD / Sim	SolidWorks, CREO Parametric, OnShape, AutoCAD, MATLAB
Mechanical	FMEA, Fatigue Analysis, S-N Curves, Goodman & Coffin-Manson Criteria, Stress Concentration Analysis, Root Cause Analysis
Languages	English (Fluent), Hindi (Native), French (Beginner)